EXPERIMENT 7

CS23334 – FUNDAMENTALS OF DATA SCIENCE LAB

Random Sampling and Sampling Distribution

Aim: Generate a population, draw samples and plot sample means.

Python Program

```
import numpy as np
import matplotlib.pyplot as plt
pop=np.random.normal(50,10,10000)
means=[]
for i in range(1000):
    means.append(np.random.choice(pop,30).mean())
plt.hist(means,bins=30)
plt.xlabel('Sample Mean'); plt.ylabel('Frequency'); plt.show()
print(pop.mean(),np.mean(means))
```

Result

The histogram displays the sampling distribution of sample means.

Note: For CSV programs, keep the named CSV file in the same folder and check that column names match.